Problem 9.13

Suppose that the effective interest rate for $\frac{1}{5}$ year is 2%. Determine the equivalent nominal interest rate, compounded every 6 years.

Solution

Let $i^{(5)}$ denote the nominal annual interest rate compounded five times per year. Since the effective interest rate for $\frac{1}{5}$ year is 2%, we have

$$\frac{i^{(5)}}{5} = 0.02.$$

Therefore,

$$i^{(5)} = 0.10.$$

Let $i^{(1/6)}$ denote the equivalent nominal annual interest rate compounded every 6 years. Since compounding every 6 years means there are $\frac{1}{6}$ conversion periods per year, the interest rate per conversion period is

$$\frac{i^{(1/6)}}{1/6} = 6i^{(1/6)}.$$

To compare the two rates over a common period, use 30 years. This is $5 \times 30 = 150$ periods of length $\frac{1}{5}$ year, and also 5 periods of length 6 years. Hence

$$\left(1 + \frac{i^{(5)}}{5}\right)^{5(30)} = \left(1 + 6i^{(1/6)}\right)^5.$$

Substituting $i^{(5)}/5 = 0.02$ gives

$$(1.02)^{150} = \left(1 + 6i^{(1/6)}\right)^5.$$

Taking the fifth root,

$$(1.02)^{30} = 1 + 6i^{(1/6)}.$$

Therefore,

$$i^{(1/6)} = \frac{(1.02)^{30} - 1}{6}.$$

Thus,

$$i^{(1/6)} \approx 0.1352269307.$$

So the equivalent nominal interest rate, compounded every 6 years, is

$$\boxed{13.52269307\%}.$$